

SkillForge

Turn Skills Into Successful Businesses.

Product & Technical Guide

A complete walkthrough of what SkillForge is, how it works, how it is built, and how to run it — the entrepreneurship enablement platform that matches your skills to real businesses and guides you from idea to first revenue.

Platform	Full-stack web application (Next.js 15 + Express + PostgreSQL)
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Document	Product & Technical Guide, Edition 1

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01 What is SkillForge?

SkillForge is an end-to-end **entrepreneurship enablement platform**. It helps everyday people — home cooks, tailors, designers, students, makers — discover a business that fits their skills and budget, learn exactly what they need, follow a step-by-step roadmap, get guidance from verified mentors, and launch their own micro-enterprise.

Think of it as a **startup accelerator operating system**: instead of a vague "go start a business", every user gets a matched idea, a learning path, a checklist, and a community — all the way to their first Rs. 10,000 in revenue.

The problem it solves. Most first-time founders in India do not lack skill — they lack a clear, trustworthy path: which business fits me, what will it cost, what do I need to learn, and what do I do first? SkillForge turns that uncertainty into a guided, measurable journey.

Who it is for

Role	What they do on the platform
Visitor	Browses business ideas, learning resources and the community without an account.
Entrepreneur	Takes a skill assessment, gets matched ideas, follows roadmaps, learns, earns certificates.
Mentor	Applies, gets verified, accepts session bookings, reviews, and runs a mentor dashboard.
Admin	Manages users, verifies mentors, approves content, handles complaints, posts announcements.

02 What You Can Do (Features)

SkillForge is organised into modules that cover the whole journey from discovery to launch.

Area	Highlights
Discovery	Skill & interest assessment, business readiness score, and a recommendation engine that ranks ideas by fit.
Business ideas	Rich detail pages: overview, market demand, SWOT, investment breakdown, licenses, marketing, risks, revenue model, lean canvas, launch checklist.
Roadmaps	Interactive 14-step journey per business with per-user progress tracking (idea validation to first customer to scale).
Learning	Video / article / PDF / template resources with lessons, enrollment, progress, and auto-issued certificates.
Mentors	Verified mentor profiles, session booking, ratings & reviews, and a mentor dashboard.
Community	Forum with success stories and questions, threaded comments, likes, and trending tags.
Dashboard	Readiness & profile rings, saved ideas, learning progress, certificates, achievements, upcoming sessions.

Area	Highlights
Notifications	In-app notifications across roadmap, learning, mentor, community, achievements and certificates.
Admin console	Analytics overview, user management, mentor verification, content approval, complaints, announcements.

03 Technology Stack

Layer	Technology	Why
Frontend	Next.js 15 (App Router), React 19, TypeScript	Modern, fast, SEO-friendly, type-safe UI.
Styling	Tailwind CSS, shadcn-style UI, Framer Motion	Consistent design system, smooth animation, dark mode.
Backend	Node.js, Express, TypeScript	Lightweight, well-understood REST API.
Database	PostgreSQL 16 + Prisma ORM	Relational integrity with a fully type-safe data layer.
Auth	JWT access + rotating refresh tokens, bcrypt, OTP, Google	Secure, stateless, revocable sessions.
Validation	Zod	Every request validated before it reaches business logic.
Media / Email	Cloudinary, Nodemailer	Offloaded uploads and transactional email.
Ops	Docker, Docker Compose	One-command local + reproducible deploys.

04 How It Is Built (Architecture)

SkillForge is a **decoupled full-stack application** split into two deployable apps that share a typed contract: a Next.js web client and an Express API, backed by PostgreSQL.

Layered (Clean) Architecture

Every backend feature is a module with the same internal layering. A request flows strictly inward, and each layer has one job:

Layer	Responsibility
Route	Declares HTTP endpoints and wires middleware (auth, RBAC, validation).
Controller	Thin. Parses the request, calls a service, shapes the response. No business logic.
Service	The heart. Owns business rules, orchestration and transactions. Never touches req / res.
Repository	The only layer that talks to Prisma. Keeps services persistence-agnostic.

Request lifecycle



Built to scale

- **Stateless API** — no server-side session, so it scales horizontally behind a load balancer.
- **Normalized database** with indexed foreign keys and pagination on every list endpoint.
- **Caching-ready** — the service layer is the natural seam to add Redis for hot reads.
- **Media offloaded** to Cloudinary; app servers stay stateless and container-friendly.

05 How It Works — User Journeys

The Entrepreneur journey

Step	What happens
1. Sign up	Create an account (email + password or Google); a 6-digit code verifies the email.
2. Assessment	Pick skills & interests, set budget, experience, available time and preferred business type.
3. Readiness score	The platform computes a 0-100 business readiness score and profile completion.
4. Recommendations	The engine ranks business ideas by how well they match the profile (see Section 6).
5. Explore an idea	Open a rich business page: investment, SWOT, licenses, marketing, revenue model, checklist.
6. Start a roadmap	Begin the 14-step launch roadmap; mark steps complete and watch progress update.
7. Learn	Enroll in short courses; completing one issues a downloadable certificate.
8. Get mentored	Browse verified mentors and book a session; leave a rating afterwards.
9. Community	Ask questions, share a success story, like and comment.

The Mentor journey

- **Apply** with a headline, areas of expertise, experience and languages — the account is upgraded to Mentor.
- **Verification** by an admin moves the profile from Pending to Verified before it appears publicly.
- **Sessions** — accept, confirm, complete or cancel booking requests from entrepreneurs.
- **Reputation** — completed sessions can be reviewed; the profile rating recomputes automatically.
- **Dashboard** — students, sessions by status, and average rating at a glance.

The Admin journey

- **Overview** — live counts of users by role, ideas, resources, mentors, sessions, posts and open complaints.
- **People** — search users, toggle active state, change roles; approve or reject pending mentors.
- **Content** — move businesses and learning resources through Draft / Review / Published / Archived.
- **Support** — triage complaints and broadcast announcements.

06 The Recommendation Engine

The engine scores every published business against the user profile and returns the top matches with a **match score from 0 to 100** and the list of matched skills. When a signed-in user asks for recommendations with no inputs, the engine reads their saved assessment automatically.

Factor	Max points	How it is scored
Skill overlap	60	Weighted fraction of the business's required skills the user has (each skill weighted).
Budget fit	25	Full points when the budget comfortably covers the range; scaled down toward the minimum; 0 if below entry cost.
Type fit	10	Matches preferred business type (home / online / offline / hybrid).
Growth potential	5	High +5, Medium +3, Low +1 — a nudge toward higher-upside ideas.

Worked example. For the demo entrepreneur "Priya" (skills: Cooking, Baking, Digital Marketing; budget Rs. 50,000; prefers a home business), the engine returns **Home Catering with a match score of 100** — full skill overlap, comfortable budget, matching type, and high growth. This was verified live against the running API.

07 Database Design

A normalized PostgreSQL schema managed with Prisma — 30+ models. The core entities:

Group	Models
Identity	User, RefreshToken, OtpToken
Profile	Skill, Interest, UserSkill, UserInterest
Catalog	Category, Business, BusinessSkill
Roadmaps	Roadmap, RoadmapStep, UserRoadmap, UserRoadmapStep
Learning	LearningResource, Lesson, Enrollment, LessonProgress, Certificate
Mentoring	MentorProfile, MentorSession, MentorReview
Community	ForumPost, Comment, PostLike
Engagement	Bookmark, Notification, Achievement, UserAchievement
Admin	Complaint, Announcement

Every table carries created / updated audit timestamps, foreign keys are indexed, and join tables that hold extra data (like skill weights or step progress) are modelled explicitly.

08 Security & Authentication

Control	Implementation
Password storage	bcrypt hashing (cost 12); OAuth-only accounts store no password.
Sessions	Short-lived JWT access token (15 min) + long-lived refresh token, rotated on every use.
Token revocation	Refresh tokens are stored server-side and can be revoked per device; reset revokes all.
Email / reset	One-time 6-digit codes (hashed, expiring) for verification and password reset.

Control	Implementation
Google login	ID token verified server-side, linked or created by Google account id.
Authorization	Role-based access control (Visitor / Entrepreneur / Mentor / Admin) on every privileged route.
Transport / headers	Helmet security headers, strict CORS allowlist, secure httpOnly cookies.
Abuse protection	Global rate limiting plus a stricter limiter on auth endpoints.
Input safety	Zod validation on every request body, query and params before controllers run.

09 API Reference (selected)

All responses use one envelope: { success, message, data, meta }. Lists include meta.pagination. Base path: /api/v1.

Module	Representative endpoints
auth	POST /auth/register · /login · /refresh · /google · /verify-otp · /forgot-password · /reset-password
users	GET /users/me/dashboard · PUT /users/me/assessment · GET/POST/DELETE /users/me/bookmarks
businesses	GET /businesses · GET /businesses/:slug · POST /businesses/recommend
roadmaps	POST /roadmaps/:id/start · GET /roadmaps/me · PATCH /roadmaps/me/steps/:id
learning	GET /learning/resources · POST /learning/resources/:id/enroll · PATCH /learning/me/lessons/:id
mentors	GET /mentors · POST /mentors/apply · POST /mentors/:id/book · POST /mentors/sessions/:id/review
community	GET /community/posts · POST /community/posts · POST /community/posts/:id/like
notifications	GET /notifications · PATCH /notifications/read-all
admin	GET /admin/overview · PATCH /admin/mentors/:id/verify · GET /admin/users

10 Running It Locally

Prerequisites: Node.js 20+, npm, and Docker (for PostgreSQL).

Quick start

```
npm install && npm run setup # root tooling + both apps
cp apps/api/.env.example apps/api/.env
cp apps/web/.env.example apps/web/.env.local
docker compose up -d db # start PostgreSQL
npm run db:deploy # create tables
npm run db:seed # demo users, ideas, mentors, roadmaps, community
```

```
npm run dev # API on :4000 + web on http://localhost:3000
```

Demo accounts (after seeding)

Role	Email	Password
Admin	admin@skillforge.app	Password123
Mentor	mentor@skillforge.app	Password123
Entrepreneur	priya@skillforge.app	Password123

Email delivery (SMTP), file uploads (Cloudinary) and Google login are wired in code and activate as soon as real credentials are added to the environment file. In development, one-time codes are printed to the API console.

11 Quality & Verification

The platform was verified live, end to end, against the running stack with real authentication tokens and seeded data.

Check	Result
Backend API smoke test	All exercised endpoints working across every module (auth, users, businesses, roadmaps, learning, mentors, community, notifications, admin).
Auth & security	Login, refresh-token rotation, RBAC blocked with 403, bad password 401, invalid input 422 — all correct.
Recommendation engine	Returns Home Catering at match score 100 for the demo profile.
Writes persist	Bookmark create/list/delete, roadmap start, community like, learning enroll — all confirmed.
Unit tests	9 / 9 passed (recommendation scoring + password hashing).
Type safety	TypeScript compiles with 0 errors on both client and server.
Production build	Next.js build succeeds for all 36 routes.

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